L33: Verification Workflow

Simulation Without a Black Box

Outline

Verification vs. Validation

Verification

Did we build the model right?

Does the simulation code faithfully implement the conceptual model?

Tool: code review, unit tests, trace debugging, analytical benchmarks.

Validation

Did we build the right model?

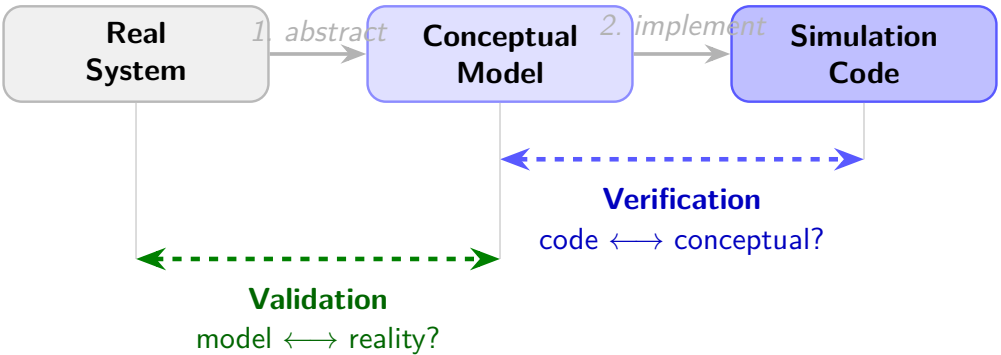
Does the conceptual model faithfully represent the real system?

Tool: historical data comparison, expert review, sensitivity checks.

Key Point

You cannot validate an unverified model — verification always comes first.

V&V Workflow



Verification must come first — you cannot validate an unverified model.

Verification Checklist

Code-level checks

- Event ordering: earlier events processed first
- Resource accounting: requests + releases balanced
- Statistics collected after warmup only
- Random seed is an explicit parameter
- No mutable global state

Behavioral checks

- Mass-balance: entities in = entities out + in-system
- Extreme conditions produce expected limits
- Single-entity trace matches hand calculation
- Results reproducible with same seed

Trace Debugging: Print State at Every Event

```
def customer(env, name, server, service_rate):
    arrive = env.now
    print(f"t={arrive:.2f}  {name} ARRIVES  queue={len(server.queue)}")
    with server.request() as req:
        yield req
        wait = env.now - arrive
        print(f"t={env.now:.2f}  {name} STARTS   wait={wait:.2f}")
        yield env.timeout(random.expovariate(service_rate))
        print(f"t={env.now:.2f}  {name} DEPARTS")
```

Rule

For every SimPy `yield`, print the simulation time, entity name, and key state variable *before and after* the yield.

Extreme-Condition Tests for M/M/1

Condition	Expected	Bug signal
$\lambda \to 0$	$\rho \to 0$, $W_q \to 0$	$\rho > 0$
$\lambda/\mu \to 1^-$	$W_q \to \infty$	finite W_q
$\rho > 1$	queue $\to \infty$	stable queue
$\mu \to \infty$	$W \to 1/\lambda$	$W \neq 1/\lambda$

Warning

$\rho > 1$ must be caught before running long experiments — an unstable simulation wastes compute and produces meaningless statistics.

Add a guard:

```
if lam / mu >= 1: raise
ValueError("rho >= 1")
```

Unit Tests: Compare to Analytical Formula

```python
import pytest
from simdes.models.queues import MM1Queue

def test_mm1_mean_wait():
    lam, mu = 0.8, 1.0          # rho = 0.8
    wq_theory = lam / (mu * (mu - lam))   # = 4.0
    q = MM1Queue(arrival_rate=lam, service_rate=mu, seed=42)
    result = q.run(sim_time=100_000)
    assert abs(result["Wq"] - wq_theory) / wq_theory < 0.05  # 5% tolerance

@pytest.mark.parametrize("lam", [0.1, 0.5, 0.9])
def test_mm1_utilisation(lam):
    q = MM1Queue(arrival_rate=lam, service_rate=1.0, seed=0)
    r = q.run(sim_time=50_000)
    assert abs(r["utilisation"] - lam) < 0.03
```

Code Review Checklist for SimPy Models

Structure

- Model is a class, not a script
- `run()` returns a dict of KPIs
- Seed passed to constructor
- No `import random` at module level

Logic

- `yield env.timeout()` never receives ≤ 0
- `Resource` capacity matches conceptual model

Statistics

- Warmup: statistics reset after warmup period
- Averages use time-weighted mean for stocks (L, Lq)
- Sample mean used for flows (W, Wq)

Common Bug

Forgetting to reset accumulators after warmup inflates means — the warm-up transient is the largest source of bias.

Common SimPy Bugs and Their Signatures

Bug	**Symptom**	**Fix**
Incorrect event ordering	Negative sojourn times	Check `env.now` ordering logic
Double-booking resource	Utilisation > 1	Ensure one `request()` per entity
Missing warmup removal	Biased mean estimates	Reset stats at warmup time
Wrong time-average formula	$L \neq \lambda W$	Use `env.now` deltas for TWMA
Shared mutable list	Results vary with seed unpredictably	Pass seed; no module-level lists

Key Takeaways

L33 Summary

1. **Verification** = code matches conceptual model; **Validation** = model matches real system.
2. Use a systematic checklist: event ordering, resource balance, warmup, reproducibility.
3. **Trace debugging** — print every event — is the fastest way to locate logic errors.
4. Extreme-condition tests ($\lambda \to 0$, $\rho \to 1^-$, $\rho > 1$) are mandatory before any experiment.
5. Automate verification with `pytest` and compare to closed-form M/M/1 or M/M/c formulas.

Next: L34 — Extreme-condition and degenerate testing in depth.